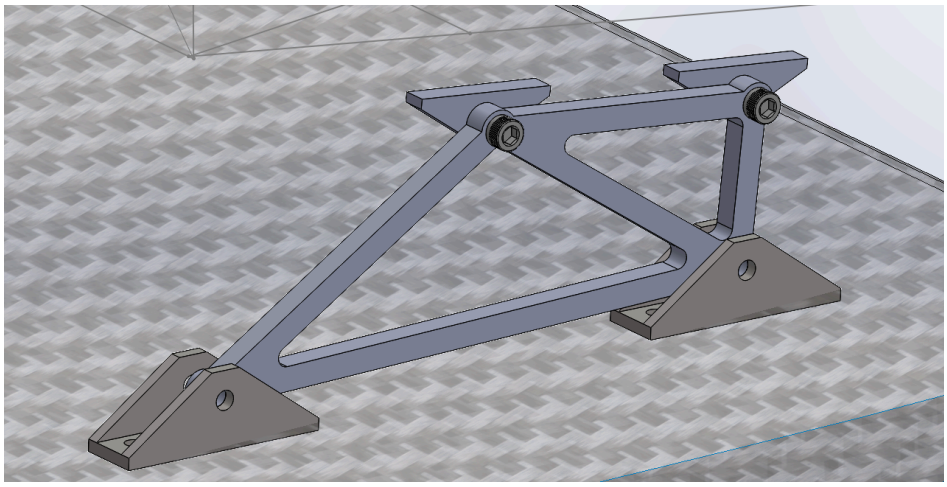
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Me 108
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ME 108 Report - FEA Study of an FSAE Front Wing Bracket



Introduction

The objective of our project is to perform a structural analysis of the front wing mounting bracket designed for the GR26, the current Formula SAE vehicle for Gaucho Racing. This specific component is a single aluminum bracket that serves as the main linkage between the main element of the front wing and the chassis nose structure. There are 2 brackets on each side of the chassis to mount the front wing, however our project just focuses on a singular bracket, and its share of the load it needs to bear. The bracket's main function is to withstand aerodynamic loads and inertial cornering forces while remaining stiff enough to ensure the front wing's angle of attack remains stable when cornering at higher speeds. We chose this object for analysis because it is critical for performance and the safety of the driver. From a performance perspective, any significant deflection would negatively impact our aero balance and from a safety perspective, the bracket must be structurally sound to meet FSAE rulebook requirements regarding controlled failure during a large impact collision or incident. By understanding the bracket's behavior under peak static loads, the team can verify that the part is not under-designed or excessively overbuilt, and we can alter designs if necessary.

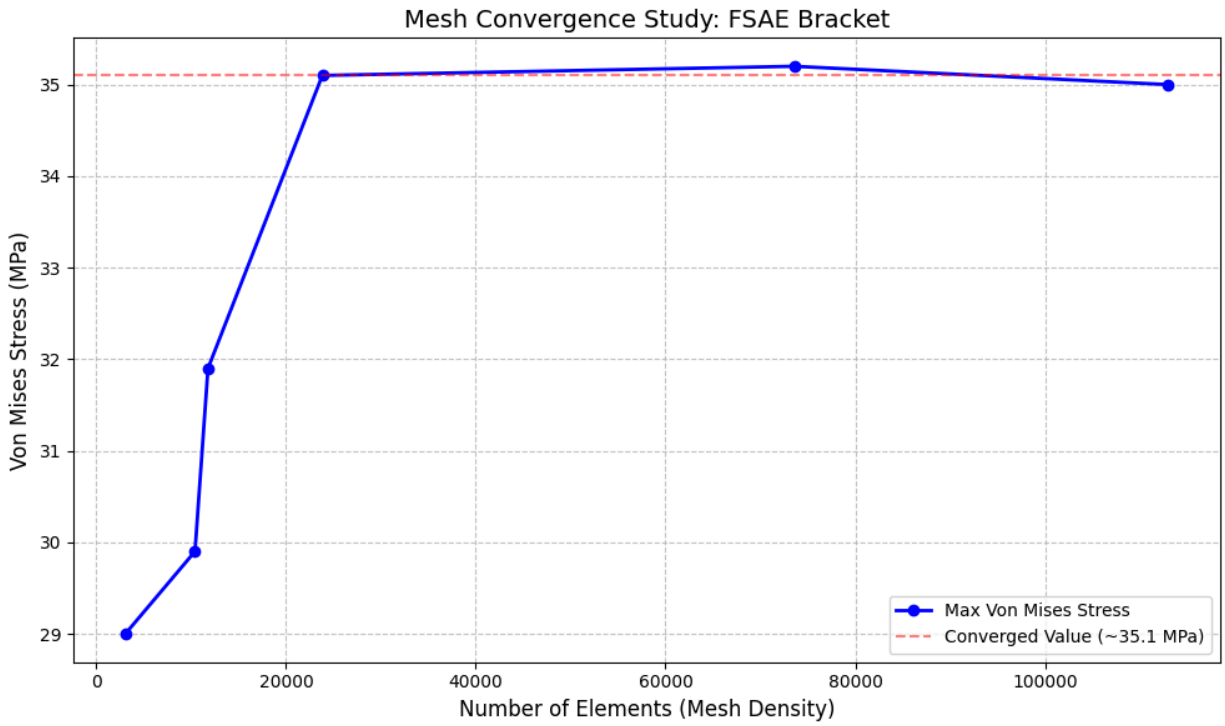
Background

Structural brackets in racing applications are traditionally studied through a lens of mass-to-stiffness optimization, as every gram saved contributes to a lower vehicle moment of inertia. Previous studies in the Formula Student community, such as research into aerodynamic system mountings, demonstrate a loading methodology consistent with this project. Specifically, this reference validates the use of simultaneous downforce and drag loads applied to fixed mounting points on their respective front wing as seen in their free body diagram and front wing assembly FEA (Bhagat).

The selection of 6061-T6 aluminum for this bracket is justified by its superior material properties compared to traditional alternatives. As noted in established modal analyses of automotive support structures, aluminum is a preferred material for brackets because "Aluminum has only about one third the density of steel and the most commercial aluminum alloys possess substantially higher specific strength than steel. A vehicle weight reduction would not only result in higher oil savings, but also gives a significant reduction in emission" (Adkine, 11). This enables the GR26 to maintain a lightweight profile without sacrificing the rigidity necessary to stabilize aerodynamic loads. By combining the loading principles found in aerodynamic mounting research with the material efficiency identified in aluminum structural studies, this report establishes a robust framework for the FEA of the front wing mounting bracket.

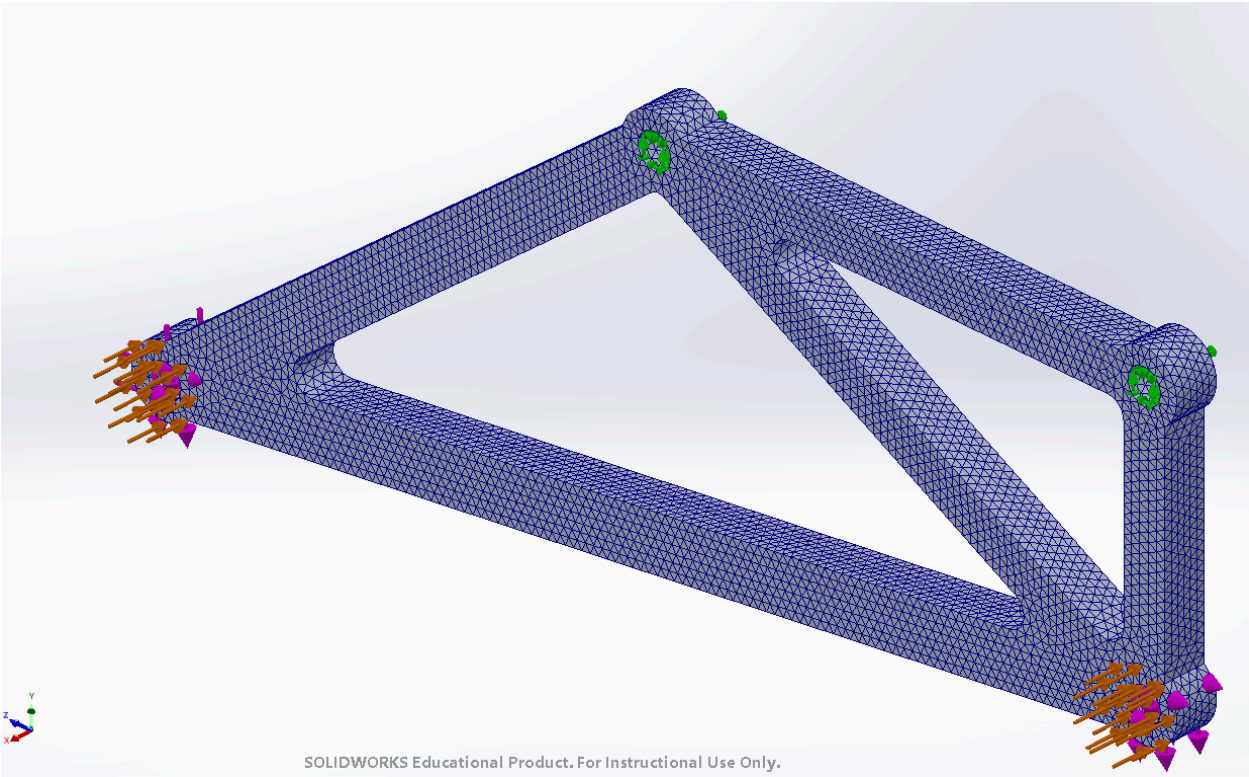
Method

The simulation was conducted using a CAD model of the GR26 bracket manufactured from 6061-T6 aluminum using SolidWorks FEA. We selected this material because of its isotropic properties, a modulus of elasticity of approximately 68.9 GPa, and a yield strength of 276 MPa. The constraints were applied as fixed supports at the holes at the top of the bracket, where the bracket bolts to the chassis trick tabs, to try and simulate a rigid connection to the vehicle frame. The loading scenario consisted of forces applied to the wing contact points, accounting for a vertical downforce of 230 N and a horizontal drag force of 46 N, representative of peak aerodynamic loads.



The mesh was generated using a curvature-based refinement strategy to ensure high fidelity around the bolt holes and geometric transitions where stress concentrations were anticipated. To ensure the accuracy and reliability of the simulation, a comprehensive mesh convergence study was performed. As shown in the convergence plot, the maximum Von Mises stress was monitored as the number of elements (mesh density) increased from approximately 5,000 to over 110,000 elements. The results demonstrated a clear convergence toward a value of approximately 35.1 MPa. Beyond 25,000 elements, the stress variation remained minimal, confirming that the solution is independent of further mesh refinement and that the results are computationally

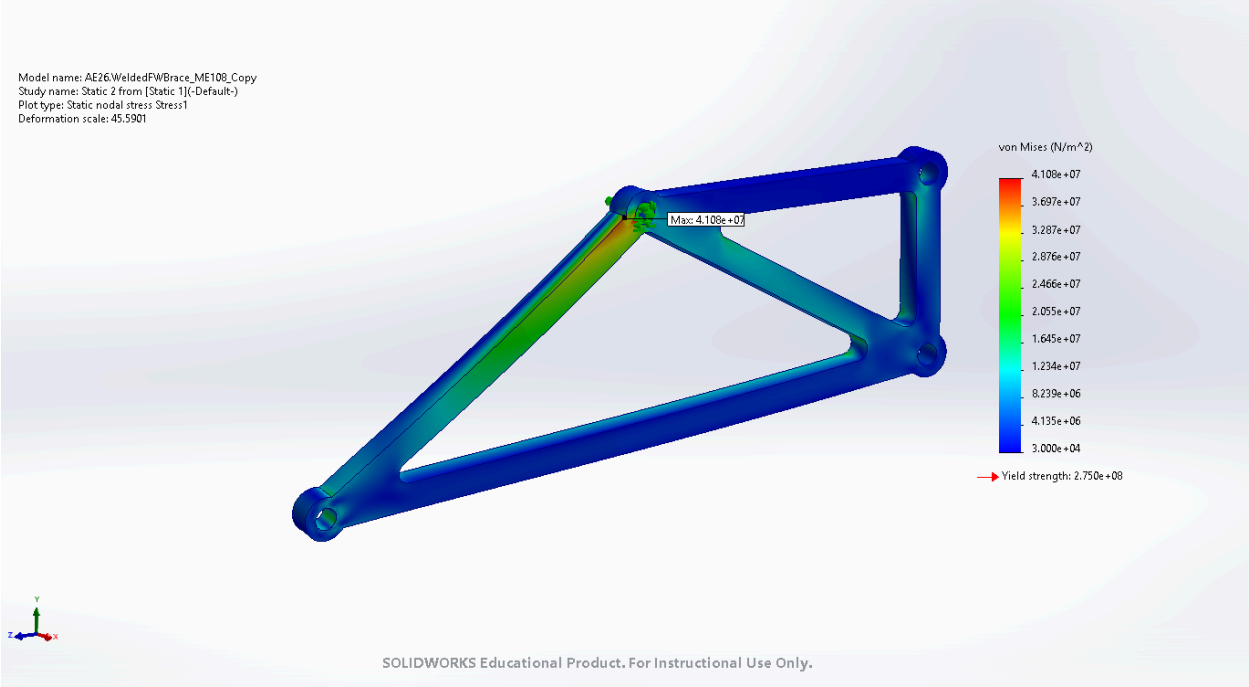
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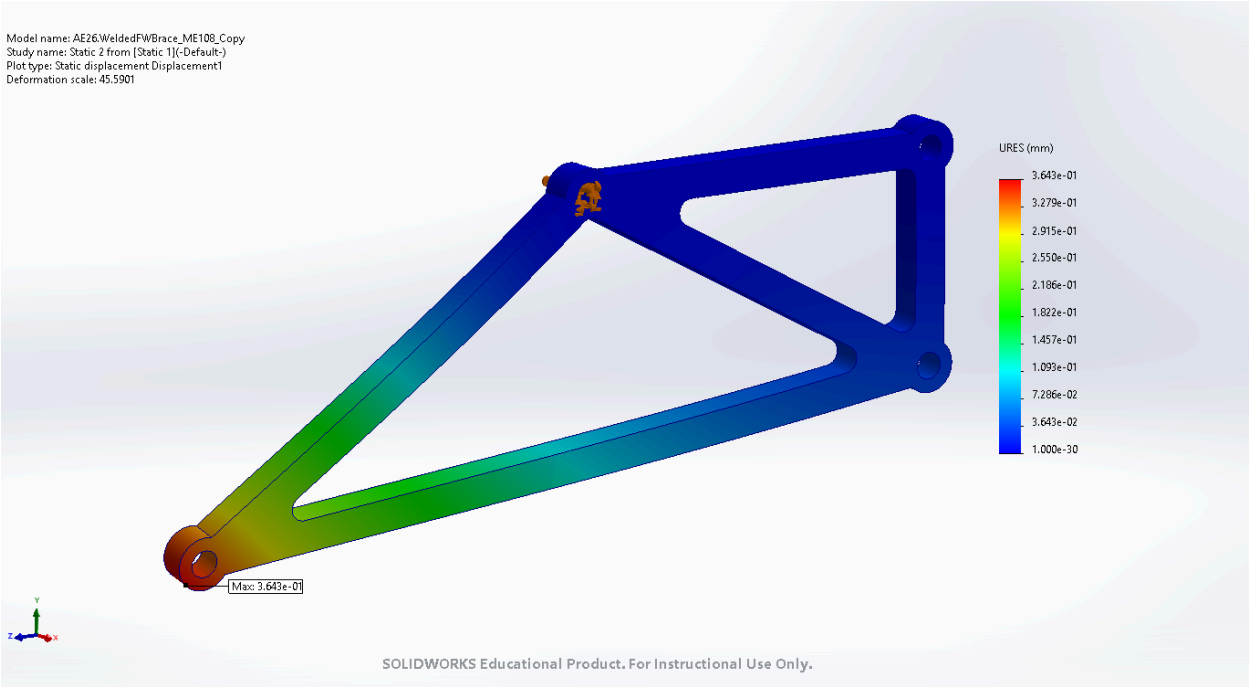
Results

The structural behavior of the GR26 bracket is visualized through stress and displacement plots. The Von Mises stress plot reveals a maximum stress of 41.08 MPa, occurring primarily at the upper mounting node where the bracket interfaces with the chassis tabs. This region experiences the highest concentration of force due to the cantilevered nature of the aerodynamic load. Notably, the yield strength of the 6061-T6 aluminum is noted at 275 MPa, indicating that

the part operates with a substantial factor of safety.

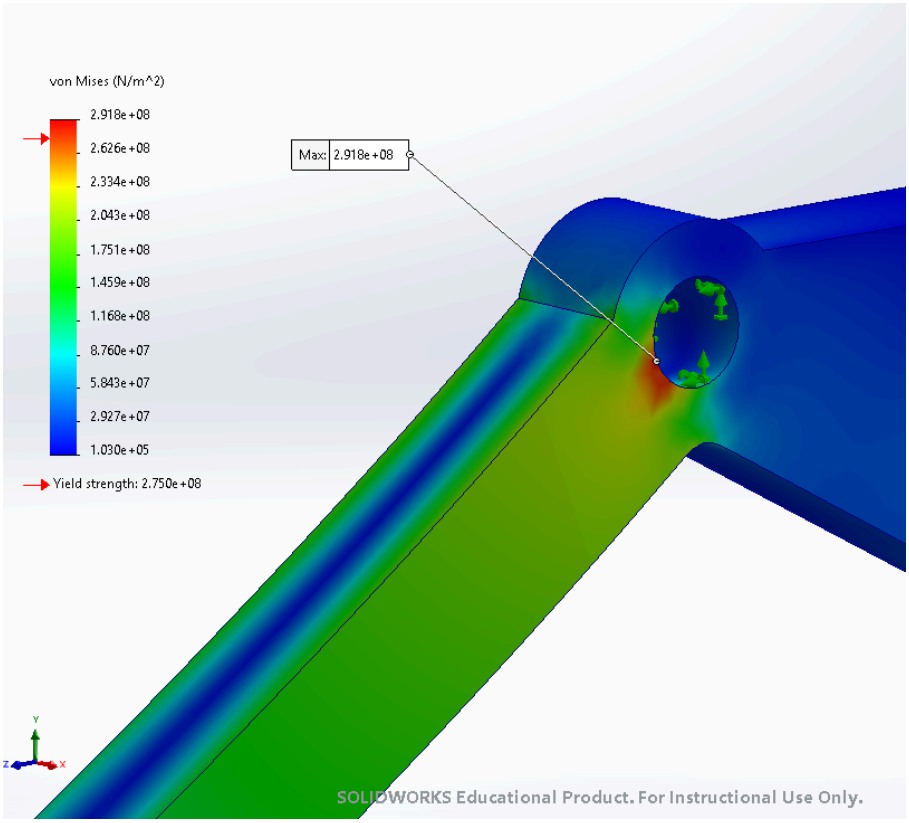


Total displacement results show that the maximum deflection occurs at the leading tip of the bracket, furthest from the fixed chassis constraints. The maximum displacement is recorded at 0.364 mm. This minimal deflection confirms that the bracket is sufficiently stiff to maintain the front wing's aerodynamic profile and angle of attack under peak static loads.

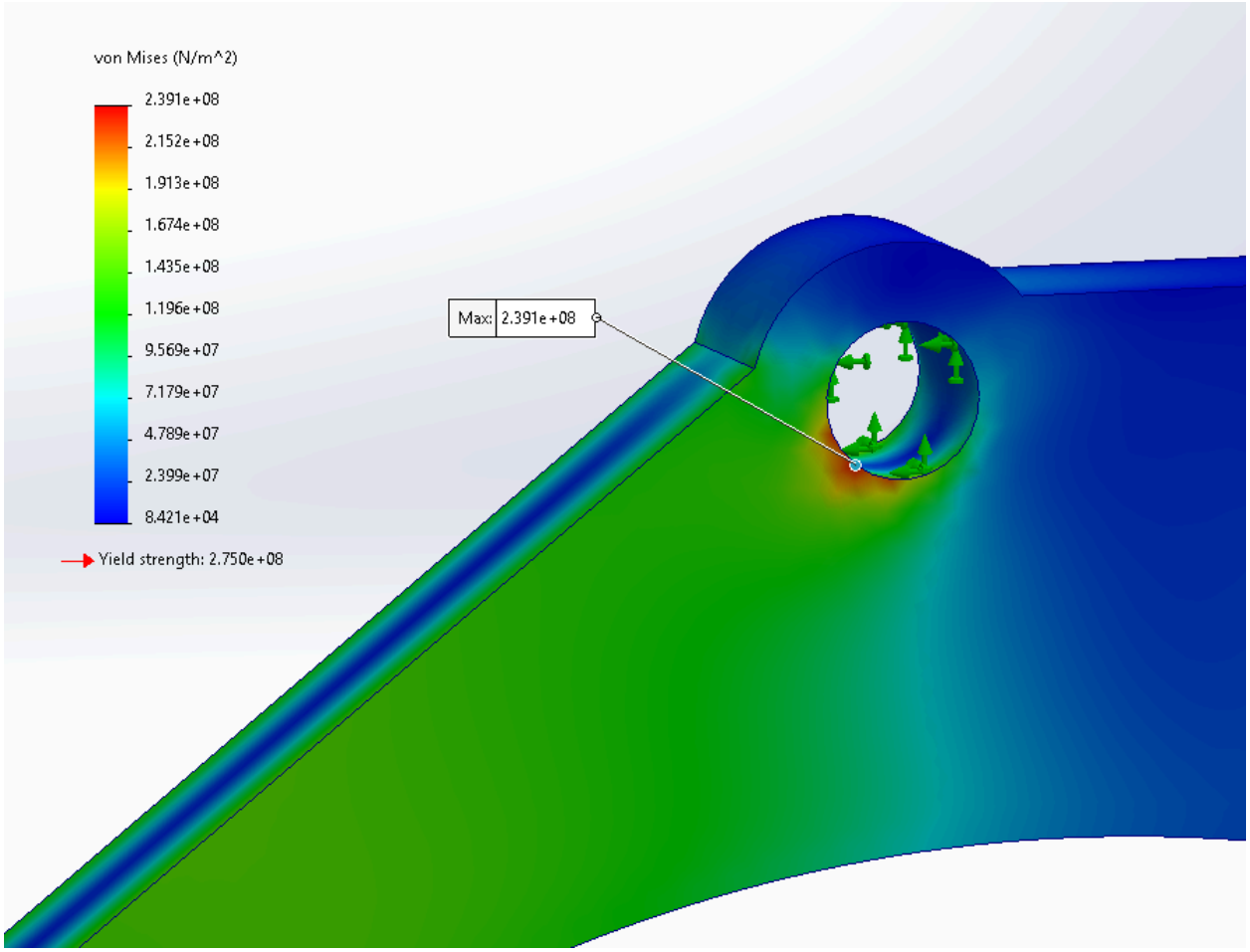


Redesign After Failure:

To get our part to fail, we increased the lateral load from 46N to 300N shared between the two loading points, essentially simulating the effect of an exaggerated horizontal cone impact. The highest stress occurred at the frontal mounting point where the part is fixed to the chassis.



There were a few factors that limited our redesign. For one, the thickness of the bracket and the hole size could not be changed since those would require an entire redesign of the entire mounting system. The next best thing we could do was to add more material near the area of high stress near the mounting point. We chose to go with a much more generous fillet of 50mm compared to the initial 3mm, which reduced maximum stress below that of failure.



Conclusion

Interpretation of the results confirms that the GR26 mounting bracket successfully meets all performance and safety requirements. The maximum recorded stress of 41.08 MPa provides a safety factor of approximately 6.7 relative to the material yield strength 275 MPa, which is well beyond the initial goal of within 50% of the yield strength. This suggests the bracket is highly structurally reliable. The displacement of 0.364 mm is negligible in the context of aerodynamic stability, ensuring the car's handling remains consistent at high speeds.

Reflecting on the project, the mesh convergence study was vital in validating that the "hot spots" at the mounting holes were converged physical stresses rather than numerical singularities. One challenge encountered was the accurate representation of the bolted interface; future work could involve a more detailed contact analysis of the double-shear configuration or

topology optimization to reduce the part's weight in areas of low stress. Overall, this experience demonstrates that the current GR26 bracket design is robust and safe for manufacturing and competition use.